Analysis of Building Frames (Using Kani's Method)

Gasper Kani, a German engineer, developed another distribution procedure based on slope deflection equations. This method is very useful for the analysis of multistorey frames. The greatest advantage of this method is, even if a mistake is committed in distribution in one of the cycles, it converges finally to the correct answer. Even today, many practising engineers who are not familiar with computer methods, use Kani's method for the analysis of 3 to 4 storey building frames.

This method is first explained for structures with fixed ends. Then, the modifications to handle simply supported and overhanging ends are discussed. Analysis of symmetric frames making use of the symmetry is also explained, after which an analysis of general frames is taken up.

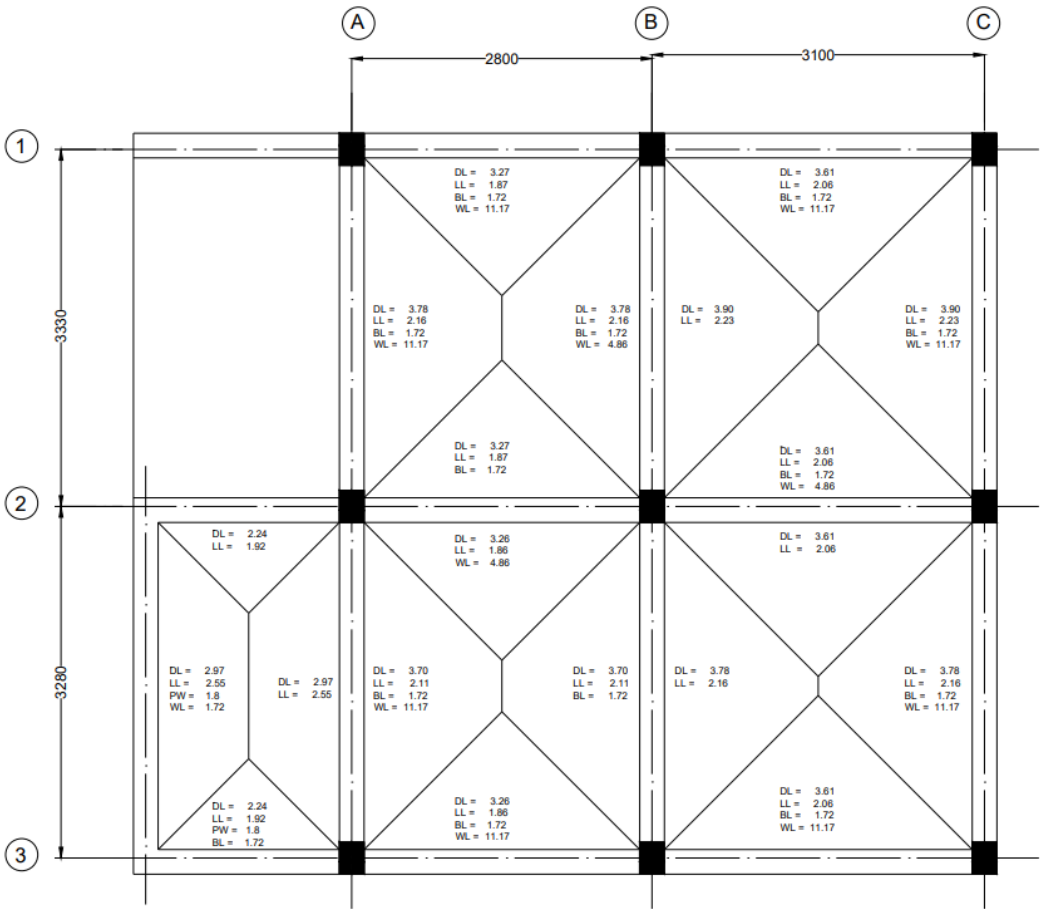


Figure 1: Beam Layout

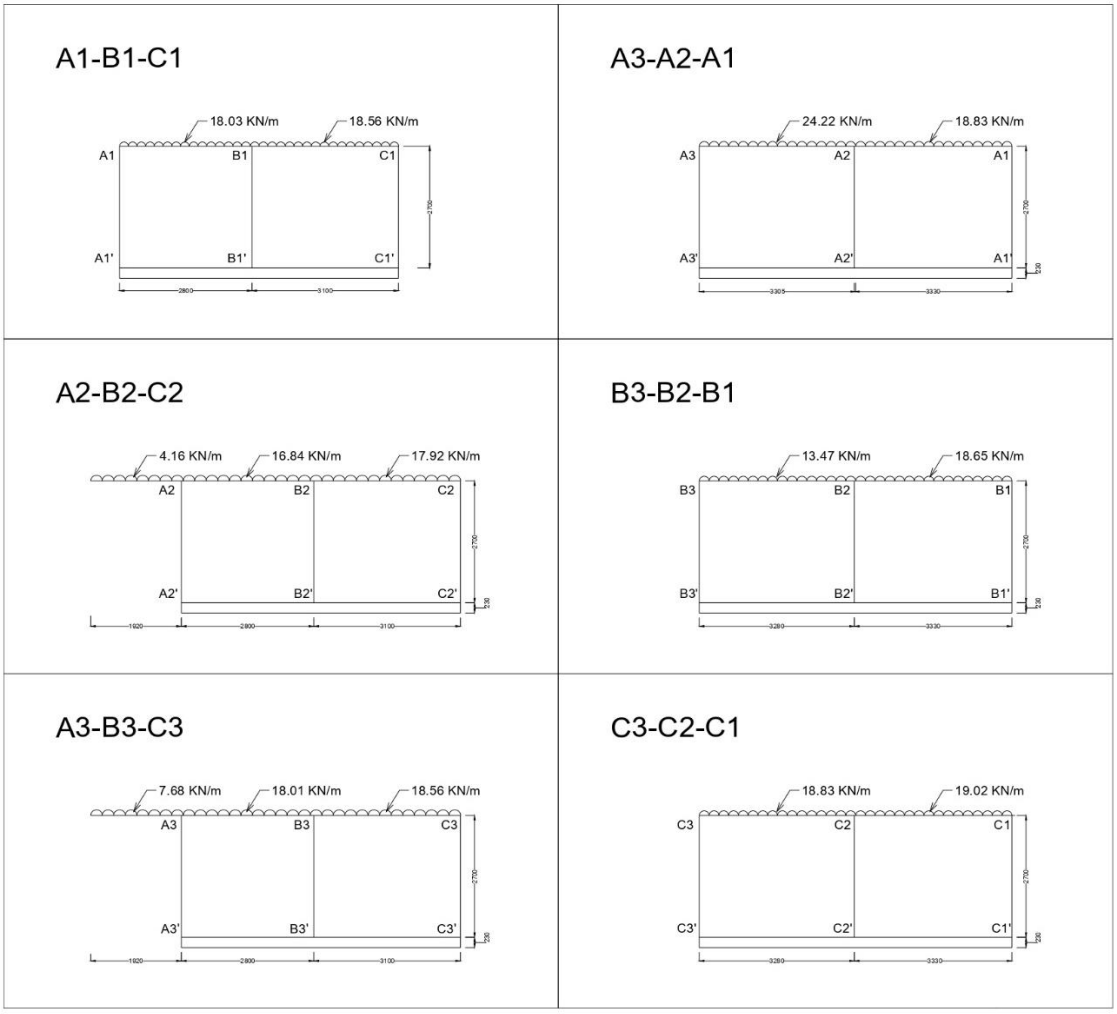
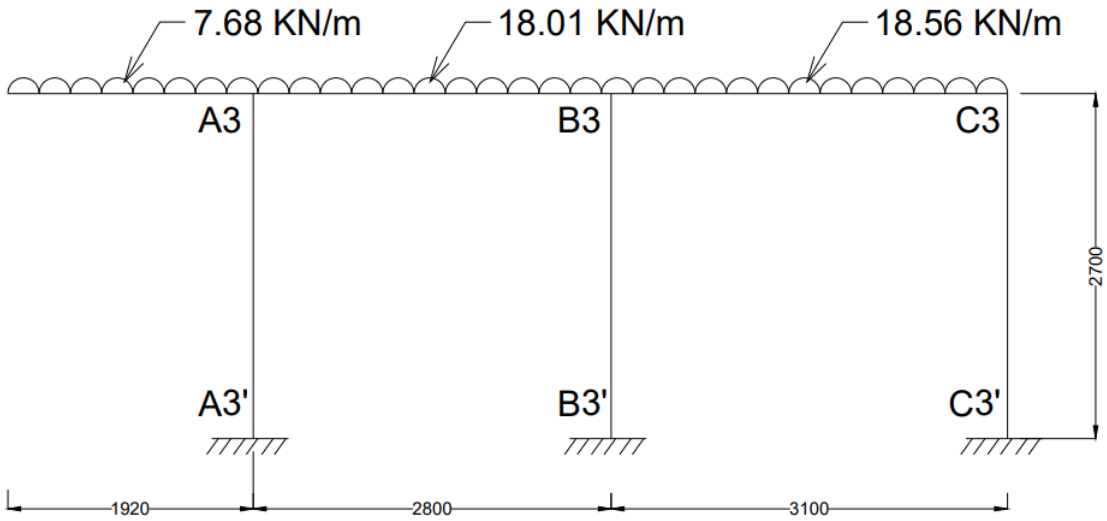


Figure 2: Frames.

Analysis of Frame A3-B3-C3



Step 1: Fixed End Moments

$MF_{Overhanging} = \frac{WL^2}{2} = \frac{7.68 \times 1.92^2}{2} = 14.16$ KN-m

$MF_{A3B3} = \frac{-WL^2}{12} = \frac{-18.01 \times 2.8^2}{12} = -11.77$ KN-m

$MF_{B3A3} = \frac{WL^2}{12} = \frac{18.01 \times 2.8^2}{12} = +11.77$ KN-m

$MF_{B3C3} = \frac{-WL^2}{12} = \frac{-18.56 \times 3.1^2}{12} = -14.86$ KN-m

$MF_{C3B3} = \frac{WL^2}{12} = \frac{18.56 \times 3.1^2}{12} = +14.86$ KN-m

Fixed end moments in All the columns are zero.

i.e,

$MF_{A3A3'} = MF_{A3'A3} = MF_{B3B3'} = MF_{B3'B3} = MF_{C3C3'} = MF_{C3'C3} = 0$

Step 2: Rotation Factor $\left(\mathbf{R.F} = -\frac{1}{2}\left(\frac{k}{\Sigma k}\right)\right), \left(k = \frac{4EI}{L}\right)$

Joints	Members	Stiffness (k)	Σk	R.F
A3	A3B3	1.428EI	2.909EI	-0.245
	A3A3′	1.481EI		-0.255
B3	B3A3	1.428EI	4.200EI	-0.170
	B3C3	1.290EI		-0.154
	B3B3′	1.481EI		-0.176
C3	C3B3	1.290EI	2.771EI	-0.233
	C3C3′	1.481EI		-0.267

Stiffness (k) $\left(k = \left(\frac{4EI}{L}\right)\right)$

$K_{A3B3} = \left(\frac{4EI}{2.8}\right) = 1.428EI$

$K_{A3A3'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

$K_{B3A3} = \left(\frac{4EI}{2.8}\right) = 1.428EI$

$K_{B3C3} = \left(\frac{4EI}{3.1}\right) = 1.290EI$

$K_{B3B3'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

$K_{C3B3} = \left(\frac{4EI}{3.1}\right) = 1.290EI$

$K_{C3C3'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

Rotation Factor $\left(\mathbf{R.F} = -\frac{1}{2}\left(\frac{k}{\Sigma k}\right)\right)$

$R.F_{A3B3} = -\frac{1}{2}\left(\frac{1.428EI}{2.909EI}\right) = -0.245$

$R.F_{A3A3'} = -\frac{1}{2}\left(\frac{1.481EI}{2.909EI}\right) = -0.255$

$R.F_{B3A3} = -\frac{1}{2}\left(\frac{1.428EI}{4.200EI}\right) = -0.170$

$R.F_{B3C3} = -\frac{1}{2}\left(\frac{1.290EI}{4.200EI}\right) = -0.154$

$R.F_{B3B3'} = -\frac{1}{2}\left(\frac{1.481EI}{4.200EI}\right) = -0.176$

$R.F_{C3B3} = -\frac{1}{2}\left(\frac{1.290EI}{2.771EI}\right) = -0.233$

$R.F_{C3C3'} = -\frac{1}{2}\left(\frac{1.481EI}{2.771EI}\right) = -0.267$

Sum of Stiffness $\sum k$ at Joints

A3: 1.428EI + 1.481EI = 2.909EI

B3: 1.428EI + 1.290EI + 1.481EI = 4.200EI

C3: 1.290EI + 1.481EI = 2.771EI

Step 3: Displacement Factor $\left(\mathbf{D.F} = -\frac{3}{2}\left(\frac{k}{\Sigma k}\right)\right)$

Storey	Members	k	Σk	D.F
1	A3A3′	1.481EI	4.444EI	-0.5
	B3B3′	1.481EI		-0.5
	C3C3′	1.481EI		-0.5

Stiffness (k) $\left(k = \left(\frac{4EI}{L}\right)\right)$

$K_{A3A3'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

$K_{B3B3'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

$K_{C3C3'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

Displacement Factor $\left(\mathbf{D.F} = -\frac{3}{2}\left(\frac{k}{\Sigma k}\right)\right)$

$D.F_{A3A3'} = -\frac{3}{2}\left(\frac{1.481EI}{4.444EI}\right) = -0.5$

$D.F_{B3B3'} = -\frac{3}{2}\left(\frac{1.481EI}{4.444EI}\right) = -0.5$

$D.F_{C3C3'} = -\frac{3}{2}\left(\frac{1.481EI}{4.444EI}\right) = -0.5$

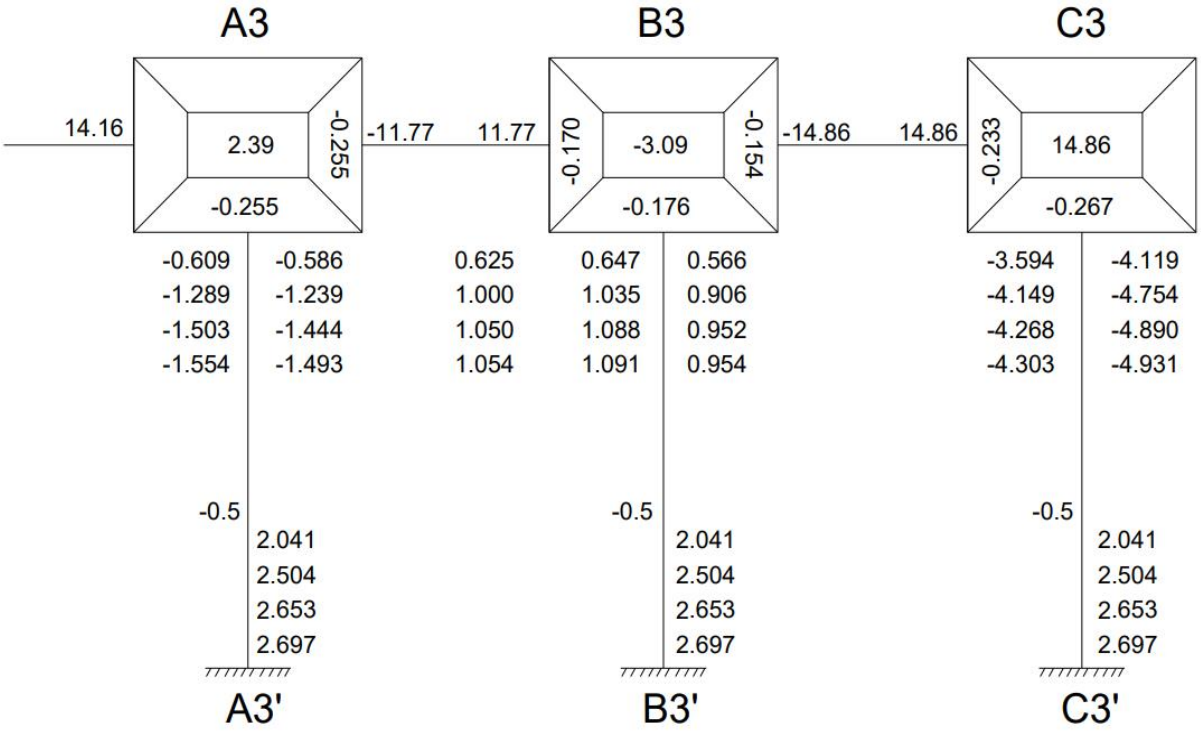
Sum of Stiffness $\sum k$ at Storey

$\sum k = 1.481EI + 1.481EI + 1.481EI = 4.444EI$

Rotation and Displacement Contribution

starting FEM at Joint A3 ∑FEM will be the sum of the overhang and the beam:

$\sum FEM_{A3} = 14.16 + (-11.77) = 2.39KN\text{-}m$



Cycle 1

Joint Rotation $(\boldsymbol{C.V.} = \boldsymbol{R.F} \times [\sum \boldsymbol{FEM} + \boldsymbol{NCV} + \boldsymbol{FCV}])$

M_{A3B3} = -0.245 × [2.39 + 0 + 0] = -0.586

$M_{A3A3'}$ = -0.255 × [2.39+ 0 + 0] = -0.609

M_{B3A3} = -0.170 × [-3.09 + (-0.586) + 0] = 0.625

M_{B3C3} = -0.154 × [-3.09 + (-0.586) + 0] = 0.566

$M_{B3B3'}$ = -0.176 × [-3.09 + (-0.586) + 0] = 0.647

M_{C3B3} = -0.233 × [14.86 + 0.566 + 0] = -3.594

$M_{C3C3'}$ = -0.267 × [14.86 + 0.566 + 0] = -4.119

Displacement Contribution

For All Columns $(\boldsymbol{R.V.} = \boldsymbol{D.F} \times [\sum \boldsymbol{Rotations\ Contribution\ at\ Columns}])$

$M_{A3A3'} + M_{B3B3'} + M_{C3C3'}$ = -0.5 × (-0.609 + 0.647 + (-4.119)) = 2.041

Cycle 2

Joint Rotation $(\boldsymbol{C.V.} = \boldsymbol{R.F} \times [\sum \boldsymbol{FEM} + \boldsymbol{NCV} + \boldsymbol{FCV}])$

M_{A3B3} = -0.245 × [2.39 + 2.041 + 0.625] = -1.239

$M_{A3A3'}$ = -0.255 × [2.39 + 2.041 + 0.625] = -1.289

M_{B3A3} = -0.170 × [-3.09 + 2.041 + (-1.239) + (-3.594)] = 1.000

M_{B3C3} = -0.154 × [-3.09 + 2.041 + (-1.239) + (-3.594)] = 0.906

$M_{B3B3'}$ = -0.176 × [-3.09 + 2.041 + (-1.239) + (-3.594)] = 1.035

M_{C3B3} = -0.233 × [14.86 + 2.041 + 0.906] = -4.149

$M_{C3C3'}$ = -0.267 × [14.86 + 2.041 + 0.906] = -4.754

Displacement Contribution

For All Columns $(\boldsymbol{R.V.} = \boldsymbol{D.F} \times [\sum \boldsymbol{Rotations\ Contribution\ at\ Columns}])$

$M_{A3A3'} + M_{B3B3'} + M_{C3C3'}$ = -0.5 × (-1.289 + 1.035 + (-4.754)) = 2.504

Cycle 3

Joint Rotation $(\boldsymbol{C.V.} = \boldsymbol{R.F} \times [\sum \boldsymbol{FEM} + \boldsymbol{NCV} + \boldsymbol{FCV}])$

M_{A3B3} = -0.245 × [2.39 + 2.504 + 1.000] = -1.444

$M_{A3A3'}$ = -0.255 × [2.39 + 2.504 + 1.000] = -1.503

M_{B3A3} = -0.170 × [-3.09 + 2.504 + (-1.444) + (-4.149)] = 1.050

M_{B3C3} = -0.154 × [-3.09 + 2.504 + (-1.444) + (-4.149)] = 0.952

$M_{B3B3'}$ = -0.176 × [-3.09 + 2.504 + (-1.444) + (-4.149)] = 1.088

M_{C3B3} = -0.233 × [14.86 + 2.504 + 0.952] = -4.268

$M_{C3C3'}$ = -0.267 × [14.86 + 2.504 + 0.952] = -4.890

Displacement Contribution

For All Columns $(\boldsymbol{R.V.} = \boldsymbol{D.F} \times [\sum \boldsymbol{Rotations\ Contribution\ at\ Columns}])$

$M_{A3A3'} + M_{B3B3'} + M_{C3C3'}$ = -0.5 × (-1.503 + 1.088 + (-4.890)) = 2.653

Cycle 4

Joint Rotation $(\boldsymbol{C.V.} = \boldsymbol{R.F} \times [\sum \boldsymbol{FEM} + \boldsymbol{NCV} + \boldsymbol{FCV}])$

M_{A3B3} = -0.245 × [2.39 + 2.653 + 1.050] = -1.493

$M_{A3A3'}$ = -0.255 × [2.39 + 2.653 + 1.050] = -1.554

M_{B3A3} = -0.170 × [-3.09 + 2.653 + (-1.493) + (-4.268)] = 1.054

M_{B3C3} = -0.154 × [-3.09 + 2.653 + (-1.493) + (-4.268)] = 0.954

$M_{B3B3'}$ = -0.176 × [-3.09 + 2.653 + (-1.493) + (-4.268)] = 1.091

M_{C3B3} = -0.233 × [14.86 + 2.653 + 0.954] = -4.303

$M_{C3C3'}$ = -0.267 × [14.86 + 2.653 + 0.954] = -4.931

Displacement Contribution

For All Columns $(\boldsymbol{R.V.} = \boldsymbol{D.F} \times [\sum \boldsymbol{Rotations\ Contribution\ at\ Columns}])$

$M_{A3A3'} + M_{B3B3'} + M_{C3C3'}$ = -0.5 × (-1.554 + 1.091 + (-4.931)) = 2.697

Step 4: Final End Moments

M_{ij} = FEM + 2×M_{ij} (Near end Rotation) + M_{ji} (Far end Rotation) +M″ (Displacement)

Beams (No Sway M″= 0) "for Joint A3, the Overhang Moment is a fixed constant of +14.16 KN-m"

M_{A3B3} = -11.77 + 2(-1.493) + 1.054 = -13.70 KN-m

M_{B3A3} = 11.77 + 2(1.054) + (-1.493) = 12.39 KN-m

M_{B3C3} = -14.86+ 2(0.954) + (-4.303) = -17.26 KN-m

M_{C3B3} = 14.86 + 2(-4.303) + 0.954 = 7.21 KN-m

Column (With Sway M″= 2.697)

$M_{A3A3'}$ = 0 + 2(-1.554) + 0 + 2.697 = -0.41 KN-m

$M_{A3'A3}$ = 0 + 2(0) + (-1.554) + 2.697 = 1.14 KN-m

$M_{B3B3'}$ = 0 + 2(1.091) + 0 + 2.697 = 4.88 KN-m

$M_{B3'B3}$ = 0 + 2(0) + 1.091 + 2.697 = 3.79 KN-m

$M_{C3C3'}$ = 0 + 2(-4.931) + 0 + 2.697 = -7.17 KN-m

$M_{C3'C3}$ = 0 + 2(0) + (-4.931) + 2.697 = -2.23 KN-m

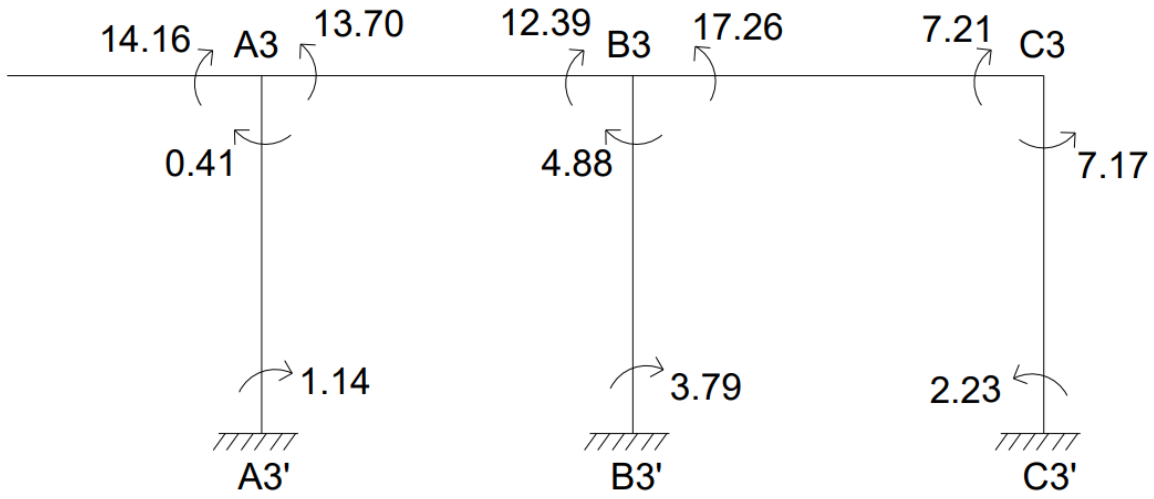
Equilibrium Check

"To check Joint A3, we must include that +14.16 overhanging moment along with the beam and column."

Joint A3 = $M_{overhanging}$ + M_{A3B3} + $M_{A3A3'}$ = 14.16 + (-13.70) + (-0.41) = 0.05 ≈ 0

Joint B3 = M_{B3A3} + $M_{B3C3'}$ + $M_{B3B3'}$ = 12.39 + (-17.26) + 4.88 = 0.01 ≈ 0

Joint C3 = M_{C3B3} + $M_{C3C3'}$ = 7.21 + (-7.17) = 0.04 ≈ 0



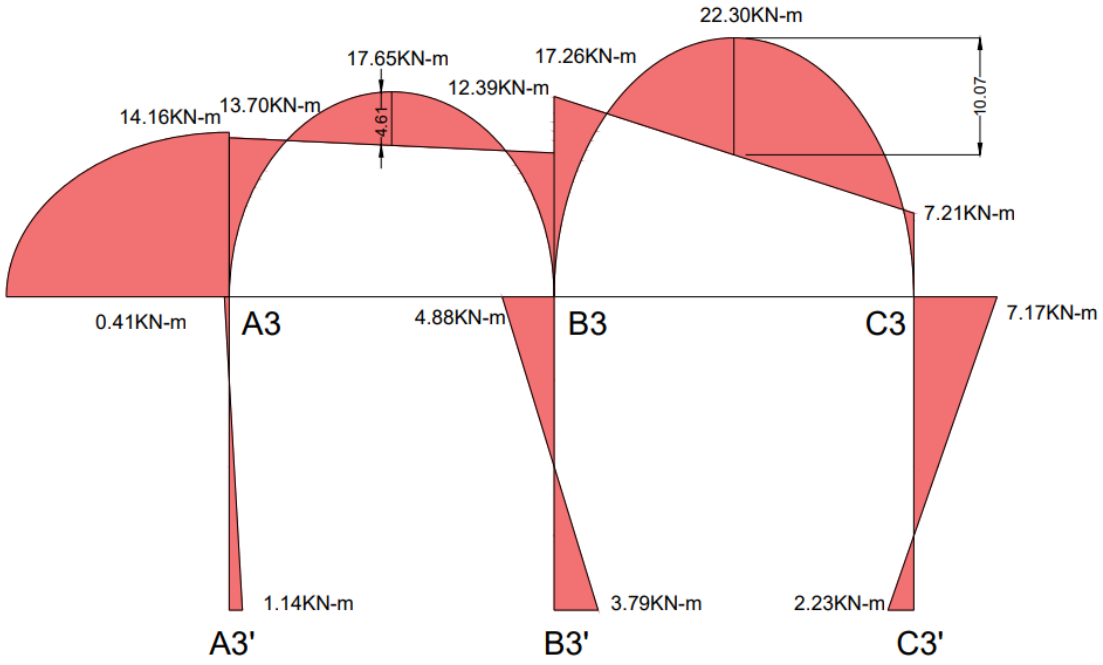
CALCULATION BENDING MOMENT

"Since the overhang is a cantilever, it doesn't have a simply supported "sagging" mid-span moment. Its moment is purely negative (hogging), starting at 0 at the free end and curving to 14.16 KN-m at support A3."

$BM_{A3B3} = \frac{WL^2}{8} = \frac{18.01 \times 2.8^2}{8} = 17.65KN\text{-}m$

$BM_{B3C3} = \frac{WL^2}{8} = \frac{18.56 \times 3.1^2}{8} = 22.30KN\text{-}m$

BENDING MOMENT DIAGRAM



CALCULATION OF REACTIONS

From Overhang (Direct load transfer):

$R_{overhang} = w \times L = 7.68 \times 1.92 = 14.75KN$

For Span A3B3

Simply Supported Reactions:

R_{total} @ A3B3$= \frac{WL}{2} = \frac{18.01 \times 2.8}{2} = 25.21KN$

Reaction Due to End Moments: (Net clockwise moment means left support gets less load)

R_{moment} @ A3B3$= \frac{M_{A3B3} - M_{B3A3}}{L} = \frac{-13.70 + 12.39}{2.8} = +0.47KN$

Reaction Due to End Moments: (Net clockwise moment means right support gets more load)

R_{moment} @ A3B3$= \frac{M_{A3B3} - M_{B3A3}}{L} = \frac{-13.70 + 12.39}{2.8} = -0.47KN$

For Span B3C3

Simply Supported Reactions:

R_{total} @ B3C3 = $\frac{WL}{2} = \frac{18.56 \times 3.1}{2}$ = 28.77KN

Reaction Due to End Moments: (Net clockwise moment means left support gets more load)

R_{moment} @ B3C3= $\frac{M_{B3C3} - M_{C3B3}}{L} = \frac{-17.26 + 7.21}{3.1}$ = +3.24KN

Reaction Due to End Moments: (Net clockwise moment means right support gets less load)

R_{moment} @ B3C3= $\frac{M_{B3C3} - M_{C3B3}}{L} = \frac{-17.26 + 7.21}{3.1}$ = -3.24KN

Final Reactions:

$R_{A3} = R_{overhang} + R_{total}$ @ A3B3 + $R_{moments}$ @ $A3B3_{(Left\ Support)}$

R_{A3} = 14.75 + 25.21 + 0.47 = 40.43KN

$R_{B3} = (R_{total}$ @ A3B3+ $R_{moments}$ @ $A3B3_{(Right\ Support)}) + (R_{total}$ @ B3C3+ $R_{moments}$ @ $B3C3_{(Left\ Support)})$ =

R_{B3} = (25.21 + (-0.47)) + (28.77 + 3.24) = 56.75KN

$R_{C3} = R_{total}$ @ B3C3+ $R_{moments}$ @ $B3C3_{(Right\ Support)}$ =

R_{C3} = 28.77 + (-3.24) = 25.53KN

